

7 Power control (PWR)

7.1 Power supplies

An internal regulator is embedded in the STM32F3xx devices.

- The internal regulator is enabled in the STM32F3xx MCUs:
The STM32F303x6/8/B/C/D/E devices require a 2.0 V - 3.6 V operating supply voltage (V_{DD}) and a 2.0 V - 3.6 V analog supply voltage (V_{DDA}). The embedded regulator is used to supply the internal 1.8 V digital power.
- The internal regulator is disabled in the STM32F3x8 MCUs:
The STM32F3x8 devices require a 1.8 V +/- 8% operating voltage supply (V_{DD}) and 1.65 V - 3.6 V analog voltage supply (V_{DDA}). The embedded regulator is OFF and V_{DD} directly supplies the regulator output.

The real-time clock (RTC) and backup registers can be powered from the V_{BAT} voltage when the main V_{DD} supply is powered off.

Figure 8. Power supply overview (STM32F303x devices)

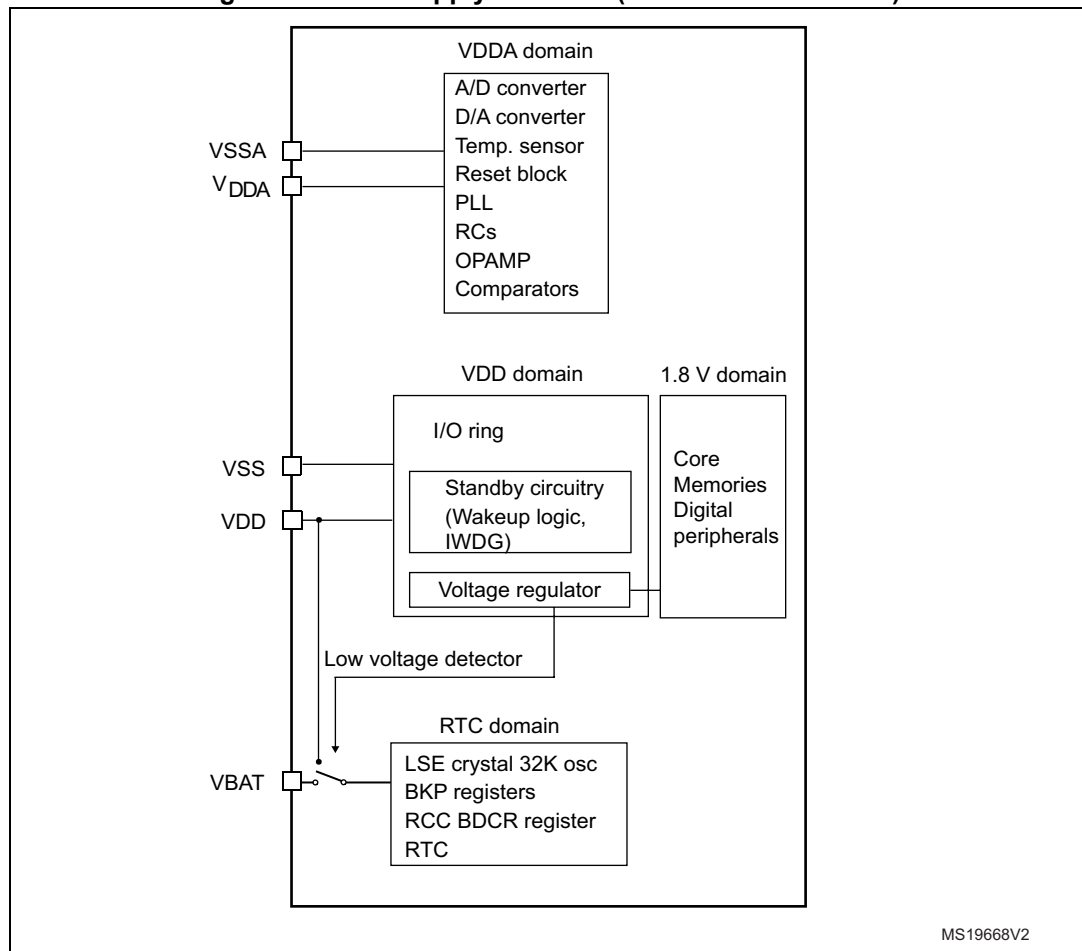
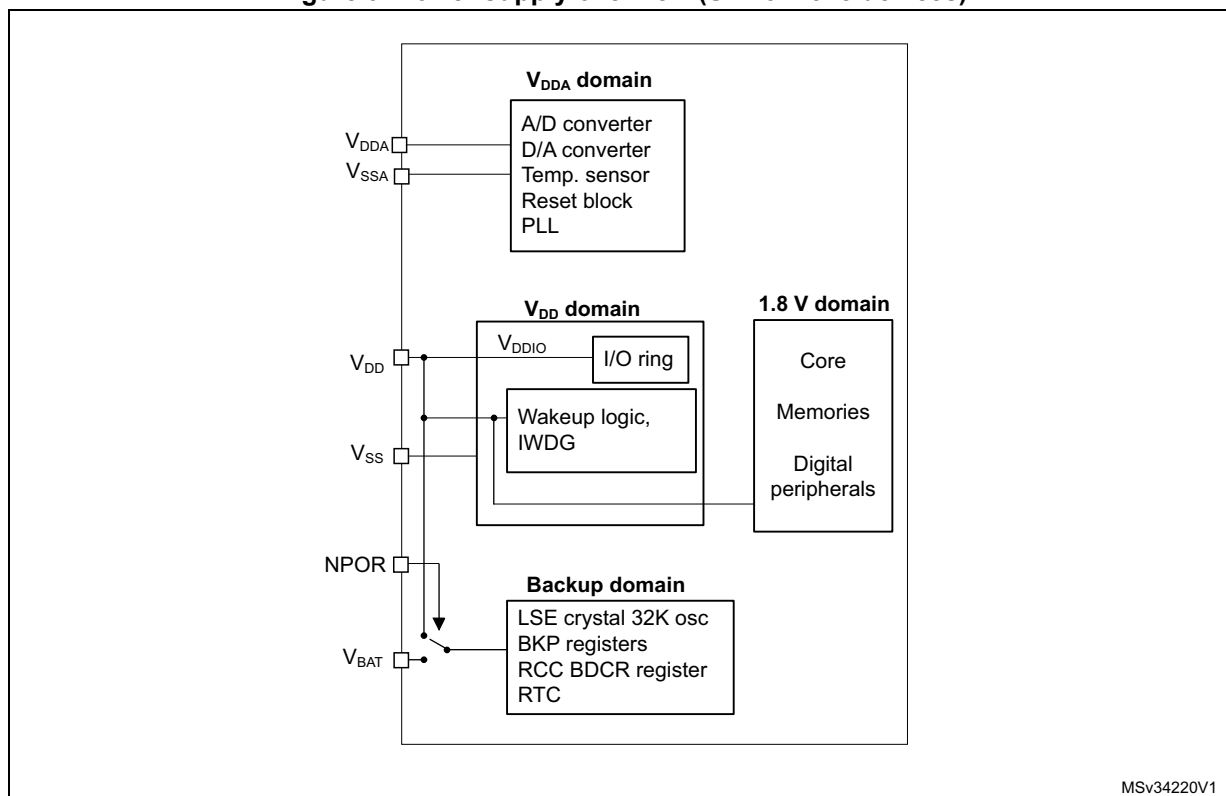


Figure 9. Power supply overview (STM32F3x8 devices)



The following supply voltages are available:

- V_{DD} and V_{SS} : external power supply for I/Os and core.
 These supply voltages are provided externally through V_{DD} and V_{SS} pins. V_{DD} = 2.0 to 3.6 V (STM32F303x6/8/B/C/D/E devices) or $1.8\text{ V} \pm 8\%$ (STM32F3x8 devices).
 When the 1.8 V mode external supply is selected, V_{DD} directly supplies the regulator output, which directly drives the VDD18 domain.
 V_{DD} must always be kept lower than or equal to V_{DDA} .
- V_{DD18} = 1.65 to 1.95 V (VDD18 domain): power supply for digital core, SRAM, and flash memory.
 V_{DD18} is either internally generated through an internal voltage regulator (STM32F303x6/8/B/C/D/E) or can be provided directly from the external V_{DD} pin when the regulator is bypassed (STM32F3x8).
- V_{DDA} , V_{SSA} = 2.0 to 3.6 V (STM32F303x6/8/B/C/D/E) or 1.65 to 3.6 V (STM32F3x8): external power supply for ADC, DAC, comparators, operational amplifiers, temperature sensor, PLL, HSI 8 MHz oscillator, LSI 40 kHz oscillator, and reset block.
 V_{DDA} must be in the 2.4 to 3.6 V range when the OPAMP and DAC are used.
 V_{DDA} must be in the 1.8 to 3.6 V range when the ADC is used.
 It is forbidden to have $V_{DDA} < V_{DD} - 0.4\text{ V}$. An external Schottky diode must be placed between V_{DD} and V_{DDA} to guarantee that this condition is met.
- V_{BAT} = 1.65 to 3.6 V: Backup power supply for RTC, LSE oscillator, PC13 to PC15 and backup registers when V_{DD} is not present. When a V_{DD} supply is present, the internal power switch switches the backup power to V_{DD} . If V_{BAT} is not used, it must be connected to V_{DD} .

7.1.1 Independent A/D and D/A converter supply and reference voltage

To improve conversion accuracy, the ADC and the DAC have an independent power supply, which can be separately filtered and shielded from noise on the PCB.

The ADC and DAC voltage supply input is available on a separate VDDA pin. An isolated supply ground connection is provided on the VSSA pin.

144- and 100-pin package connections

To ensure a better accuracy on low-voltage inputs and outputs, a separate external reference voltage can be connected on VREF+. VREF+ is the highest voltage, represented by the full-scale value, for an analog input (ADC) or output (DAC) signal.

64-pin, 48-pin, and 32-pin package connections

On these packages, the VREF+ and VREF- pins are not available. They are internally connected to the ADC voltage supply (V_{DDA}) and ground (V_{SSA}) respectively.

The V_{DDA} supply/reference voltage can be equal to or higher than V_{DD} . When a single supply is used, V_{DDA} can be externally connected to V_{DD} , through the external filtering circuit to ensure a noise free V_{DDA} /reference voltage.

When V_{DDA} is different from V_{DD} , V_{DDA} must always be higher or equal to V_{DD} . To maintain a safe potential difference between V_{DDA} and V_{DD} during power-up/power-down, an external Schottky diode can be used between V_{DD} and V_{DDA} . Refer to the datasheet for the maximum allowed difference.

7.1.2 Battery backup domain

To retain the content of the backup registers and supply the RTC function when V_{DD} is turned off, V_{BAT} pin can be connected to an optional standby voltage supplied by a battery or another source.

The V_{BAT} pin powers the RTC unit, the LSE oscillator and the PC13 to PC15 I/Os, allowing the RTC to operate even when the main power supply is turned off. The switch to the V_{BAT} supply is controlled by the power-down reset (PDR) embedded in the reset block.

Warning: During $t_{RSTTEMPO}$ (temporization at V_{DD} startup) or after a PDR is detected, the power switch between V_{BAT} and V_{DD} remains connected to V_{BAT} .
 During the startup phase, if V_{DD} is established in less than $t_{RSTTEMPO}$ (refer to the datasheet for the value of $t_{RSTTEMPO}$) and $V_{DD} > V_{BAT} + 0.6$ V, a current may be injected into V_{BAT} through an internal diode connected between V_{DD} and the power switch (V_{BAT}).
 If the power supply/battery connected to the V_{BAT} pin cannot support this current injection, it is strongly recommended to connect an external low-drop diode between this power supply and the V_{BAT} pin.

If no external battery is used in the application, it is recommended to connect V_{BAT} to the V_{DD} supply and add a 100 nF ceramic decoupling capacitor on the V_{BAT} pin. For more details, refer to AN4206.

When the RTC domain is supplied by V_{DD} (analog switch connected to V_{DD}), the following functions are available:

- PC13, PC14, and PC15 can be used as GPIO pins
- PC13, PC14, and PC15 can be configured by RTC or LSE (refer to [Section 27.3: RTC functional description on page 788](#))

Note: Because the switch only sinks a limited amount of current (3 mA), the use of GPIOs PC13 to PC15 in output mode is restricted: the speed has to be limited to 2 MHz with a maximum load of 30 pF and these I/Os must not be used as a current source (for example, to drive an LED).

When the RTC domain is supplied by V_{BAT} (analog switch connected to V_{BAT} because V_{DD} is not present), the following functions are available:

- PC13, PC14, and PC15 can be controlled only by RTC or LSE (refer to [Section 27.3: RTC functional description on page 788](#))

7.1.3 Voltage regulator

The voltage regulator is always enabled after Reset. It works in three different modes depending on the application modes.

- In Run mode, the regulator supplies full power to the 1.8 V domain (core, memories, and digital peripherals).
- In Stop mode the regulator supplies low-power to the 1.8 V domain, preserving the contents of registers and SRAM.
- In Standby Mode, the regulator is powered off. The contents of the registers and SRAM are lost except for the Standby circuitry and the RTC Domain.

In the STM32F3x8 devices, the voltage regulator is bypassed and the microcontroller must be powered from a nominal $V_{DD} = 1.8 \text{ V} \pm 8\%$ voltage.

7.2 Power supply supervisor

7.2.1 Power on reset (POR)/power down reset (PDR)

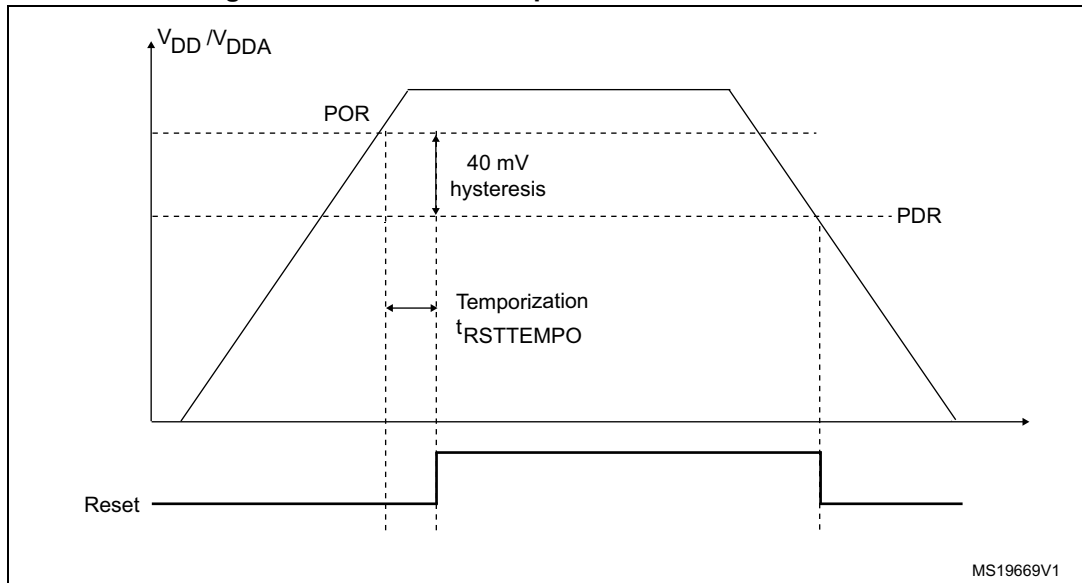
The device has an integrated power-on reset (POR) and power-down reset (PDR) circuits, which are always active and ensure proper operation above a threshold of 2 V.

The device remains in Reset mode when the monitored supply voltage is below a specified threshold, $V_{POR/PDR}$, without the need for an external reset circuit.

- The POR monitors only the V_{DD} supply voltage. During the startup phase V_{DDA} must arrive first and be greater than or equal to V_{DD} .
- The PDR monitors both the V_{DD} and V_{DDA} supply voltages. However, if the application is designed with V_{DDA} higher than or equal to V_{DD} , the V_{DDA} power supply supervisor can be disabled (by programming a dedicated $VDDA_MONITOR$ option bit) to reduce the power consumption.

For more details on the power on /power down reset threshold, refer to the electrical characteristics section in the datasheet.

Figure 10. Power on reset/power down reset waveform



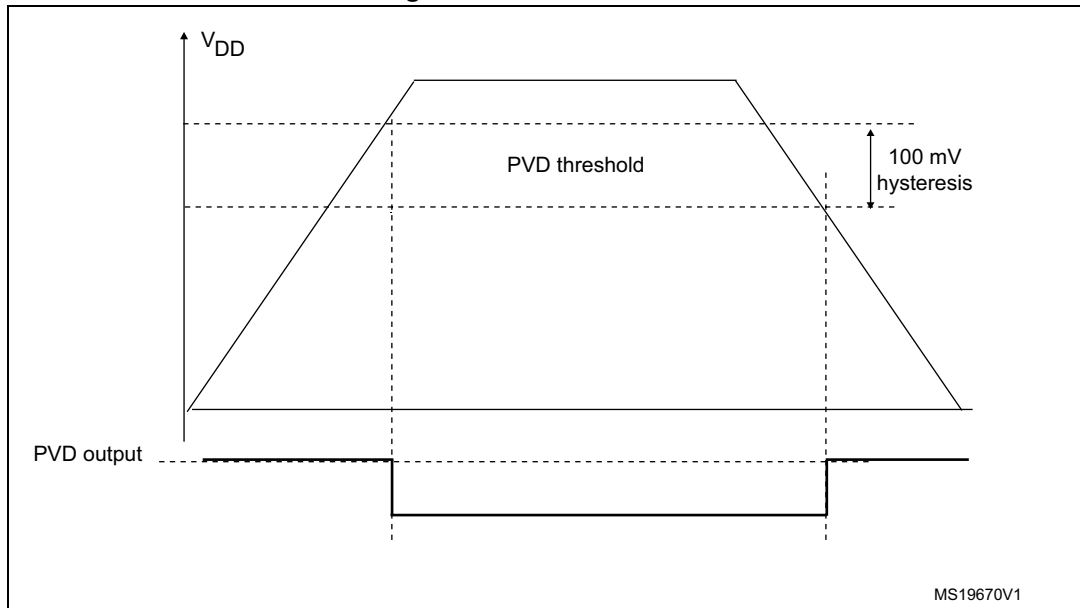
7.2.2 Programmable voltage detector (PVD)

User can use the PVD to monitor the V_{DD} power supply by comparing it to a threshold selected by the PLS[2:0] bits in the [Power control register \(PWR_CR\)](#).

The PVD is enabled by setting the PVDE bit.

A PVDO flag is available, in the [Power control/status register \(PWR_CSR\)](#), to indicate if V_{DD} is higher or lower than the PVD threshold. This event is internally connected to the EXTI line16 and can generate an interrupt if enabled through the EXTI registers. The rising/falling edge sensitivity of the EXTI Line16 should be configured according to PVD output behavior that is, if the EXTI line 16 is configured to rising edge sensitivity, the interrupt is generated when V_{DD} drops below the PVD threshold. As an example the service routine could perform emergency shutdown tasks.

Figure 11. PVD thresholds



Note: In the STM32F3x8 devices ($V_{DD} = 1.8\text{ V} \pm 8\%$), the POR, PDR, and PVD features are not available. The Power on reset signal is applied on the NPOR pin. See details in the following section.

7.2.3 External NPOR signal

In the STM32F3x8 devices, the PB2 I/O is not available and is replaced by the NPOR functionality used for Power-on reset.

To guarantee a proper power on reset, the NPOR pin must be held low when V_{DDA} is applied. When V_{DD} is stable, the reset state can be exited either by:

- putting the NPOR pin in high impedance. NPOR pin has an internal pull-up, which holds this input to V_{DDA}
- or forcing the pin to a high level by connecting it externally to V_{DDA} through a pull-up resistor.

7.3 Low-power modes

By default, the microcontroller is in Run mode after a system or a power Reset. Several low-power modes are available to save power when the CPU does not need to be kept running, for example when waiting for an external event. It is up to the user to select the mode that gives the best compromise between low-power consumption, short startup time and available wake-up sources.

The device features three low-power modes:

- Sleep mode (CPU clock off, all peripherals including Arm® Cortex®-M4 core peripherals like NVIC, SysTick, etc. are kept running)
- Stop mode (all clocks are stopped)
- Standby mode (1.8V domain powered-off)

In addition, the power consumption in Run mode can be reduced by one of the following means:

- Slowing down the system clocks
- Gating the clocks to the APB and AHB peripherals when they are unused.

Table 17. Low-power mode summary

Mode name	Entry	wakeup	Effect on 1.8V domain clocks	Effect on V _{DD} domain clocks	Voltage regulator
Sleep (Sleep now or Sleep-on - exit)	WFI	Any interrupt	CPU clock OFF no effect on other clocks or analog clock sources	None	ON
	WFE	Wakeup event			
Stop	PDDS and LPDS bits + SLEEPDEEP bit + WFI or WFE	Any EXTI line (configured in the EXTI registers) Specific communication peripherals on reception events (USART, I2C)	All 1.8V domain clocks OFF	HSI and HSE oscillators OFF	ON or in low-power mode (depends on Power control register (PWR_CR))
Standby	PDDS bit + SLEEPDEEP bit + WFI or WFE	WKUP pin rising edge, RTC alarm, external reset in NRST pin, IWDG reset			OFF

Caution: In STM32F3x8 devices with regulator off, Standby mode is not available. Stop mode is still available but it is meaningless to distinguish between voltage regulator in low-power mode and voltage regulator in Run mode because the regulator is not used and V_{DD} is applied externally to the regulator output.

7.3.1 Slowing down system clocks

In Run mode the speed of the system clocks (SYSCLK, HCLK, PCLK) can be reduced by programming the prescaler registers. These prescalers can also be used to slow down peripherals before entering Sleep mode.

For more details, refer to [Section 9.4.2: Clock configuration register \(RCC_CFGR\)](#).

7.3.2 Peripheral clock gating

In Run mode, the HCLK, and PCLK for individual peripherals and memories can be stopped at any time to reduce power consumption.

To further reduce power consumption in Sleep mode the peripheral clocks can be disabled prior to executing the WFI or WFE instructions.

Peripheral clock gating is controlled by the AHB peripheral clock enable register (RCC_AHBENR), APB1 peripheral clock enable register (RCC_APB1ENR) and APB2 peripheral clock enable register (RCC_APB2ENR).

7.3.3 Sleep mode

Entering Sleep mode

The Sleep mode is entered by executing the WFI (Wait For Interrupt) or WFE (Wait for Event) instructions. Two options are available to select the Sleep mode entry mechanism, depending on the SLEEPONEXIT bit in the Arm[®] Cortex[®]-M4 System Control register:

- Sleep-now: if the SLEEPONEXIT bit is cleared, the MCU enters Sleep mode as soon as the WFI or WFE instruction is executed.
- Sleep-on-exit: if the SLEEPONEXIT bit is set, the MCU enters Sleep mode as soon as it exits the lowest priority ISR.

In the Sleep mode, all I/O pins keep the same state as in the Run mode.

Refer to [Table 18](#) and [Table 19](#) for details on how to enter Sleep mode.

Exiting Sleep mode

If the WFI instruction is used to enter Sleep mode, any peripheral interrupt acknowledged by the nested vectored interrupt controller (NVIC) can wake up the device from Sleep mode.

If the WFE instruction is used to enter Sleep mode, the MCU exits Sleep mode as soon as an event occurs. The wake-up event can be generated either by:

- enabling an interrupt in the peripheral control register but not in the NVIC, and enabling the SEVONPEND bit in the Cortex-M4 System Control register. When the MCU resumes from WFE, the peripheral interrupt pending bit and the peripheral NVIC IRQ channel pending bit (in the NVIC interrupt clear pending register) have to be cleared.
- or configuring an external or internal EXTI line in event mode. When the CPU resumes from WFE, it is not necessary to clear the peripheral interrupt pending bit or the NVIC IRQ channel pending bit as the pending bit corresponding to the event line is not set.

This mode offers the lowest wake-up time as no time is wasted in interrupt entry/exit.

Refer to [Table 18](#) and [Table 19](#) for more details on how to exit Sleep mode.

Table 18. Sleep-now

Sleep-now mode	Description
Mode entry	WFI (Wait for Interrupt) or WFE (Wait for Event) while: – SLEEPDEEP = 0 and – SLEEPONEXIT = 0 Refer to the Cortex-M4 System Control register.
Mode exit	If WFI was used for entry: Interrupt: Refer to Table 82: STM32F303xB/C/D/E, STM32F358xC and STM32F398xE vector table and Table 83: STM32F303x6/8 and STM32F328x8 vector table . If WFE was used for entry: Wakeup event: Refer to Section 14.2.3: Wake-up event management
Wakeup latency	None

Table 19. Sleep-on-exit

Sleep-on-exit	Description
Mode entry	WFI (wait for interrupt) while: – SLEEPDEEP = 0 and – SLEEPONEXIT = 1 Refer to the Cortex-M4 System Control register.
Mode exit	Interrupt: refer to Table 82: STM32F303xB/C/D/E, STM32F358xC and STM32F398xE vector table and Table 83: STM32F303x6/8 and STM32F328x8 vector table .
Wakeup latency	None

7.3.4 Stop mode

The Stop mode is based on the Cortex-M4 deep-sleep mode combined with peripheral clock gating. The voltage regulator can be configured either in normal or low-power mode in the STM32F3xx devices. In the STM32F3x8 devices, it is meaningless to distinguish between voltage regulator in low-power mode and voltage regulator in Run mode because the regulator is not used and V_{DD} is applied externally to the regulator output. In Stop mode, all clocks in the 1.8 V domain are stopped, the PLL, the HSI, and the HSE RC oscillators are disabled. SRAM and register contents are preserved.

In the Stop mode, all I/O pins keep the same state as in the Run mode.

Entering Stop mode

Refer to [Table 20](#) for details on how to enter the Stop mode.

To further reduce power consumption in Stop mode, the internal voltage regulator can be put in low-power mode. This is configured by the LPDS bit of the [Power control register \(PWR_CR\)](#).

If flash memory programming is ongoing, the Stop mode entry is delayed until the memory access is finished.

If an access to the APB domain is ongoing, The Stop mode entry is delayed until the APB access is finished.

In Stop mode, the following features can be selected by programming individual control bits:

- Independent watchdog (IWDG): the IWDG is started by writing to its Key register or by hardware option. Once started, it cannot be stopped except by a Reset. See [Section 25.3: IWDG functional description](#) in [Section 25: Independent watchdog \(IWDG\)](#).
- real-time clock (RTC): this is configured by the RTCEN bit in the [RTC domain control register \(RCC_BDCR\)](#)
- Internal RC oscillator (LSI RC): this is configured by the LSION bit in the [Control/status register \(RCC_CSR\)](#).
- External 32.768 kHz oscillator (LSE OSC): this is configured by the LSEON bit in the [RTC domain control register \(RCC_BDCR\)](#).

The ADC or DAC can also consume power during the Stop mode, unless they are disabled before entering it. To disable the ADC, the ADDIS bit must be set in the ADCx_CR register. To disable the DAC, the ENx bit in the DAC_CR register must be written to 0.

Exiting Stop mode

Refer to [Table 20](#) for more details on how to exit Stop mode.

When exiting Stop mode by issuing an interrupt or a wake-up event, the HSI RC oscillator is selected as the system clock.

When the voltage regulator operates in low-power mode, an additional startup delay is incurred when waking up from Stop mode. By keeping the internal regulator ON during Stop mode, the consumption is higher although the startup time is reduced.

Table 20. Stop mode

Stop mode	Description
Mode entry	WFI (Wait for Interrupt) or WFE (Wait for Event) while: – Set SLEEPDEEP bit in Arm® Cortex®-M4 System Control register – Clear PDDS bit in Power Control register (PWR_CR) – Select the voltage regulator mode by configuring LPDS bit in PWR_CR Note: To enter Stop mode, all EXTI Line pending bits (in Pending register (EXTI_PR1)), all peripherals interrupt pending bits and RTC Alarm flag must be reset. Otherwise, the Stop mode entry procedure is ignored and program execution continues. If the application needs to disable the external oscillator (external clock) before entering Stop mode, the system clock source must be first switched to HSI and then clear the HSEON bit. Otherwise, if before entering Stop mode the HSEON bit is kept at 1, the security system (CSS) feature must be enabled to detect any external oscillator (external clock) failure and avoid a malfunction when entering Stop mode.
Mode exit	If WFI was used for entry: – Any EXTI Line configured in Interrupt mode (the corresponding EXTI Interrupt vector must be enabled in the NVIC). – Some specific communication peripherals (USART, I2C) interrupts, when programmed in wakeup mode (the peripheral must be programmed in wakeup mode and the corresponding interrupt vector must be enabled in the NVIC). Refer to Table 82: STM32F303xB/C/D/E, STM32F358xC and STM32F398xE vector table and Table 83: STM32F303x6/8 and STM32F328x8 vector table. If WFE was used for entry: Any EXTI Line configured in event mode. Refer to Section 14.2.3: Wake-up event management
Wakeup latency	HSI RC wakeup time + regulator wakeup time from Low-power mode

7.3.5 Standby mode

The Standby mode allows to achieve the lowest power consumption. It is based on the Cortex-M4 deepsleep mode, with the voltage regulator disabled. The 1.8 V domain is consequently powered off. The PLL, the HSI oscillator and the HSE oscillator are also switched off. SRAM and register contents are lost except for registers in the RTC domain and Standby circuitry (see [Figure 8](#)).

Caution: In the STM32F3x8 devices, the Standby mode is not available.

Entering Standby mode

Refer to [Table 21](#) for more details on how to enter Standby mode.

In Standby mode, the following features can be selected by programming individual control bits:

- Independent watchdog (IWDG): the IWDG is started by writing to its Key register or by hardware option. Once started, it cannot be stopped except by a reset. See [Section 25.3: IWDG functional description](#) in [Section 25: Independent watchdog \(IWDG\)](#).
- real-time clock (RTC): this is configured by the RTCEN bit in the RTC domain control register (RCC_BDCR)
- Internal RC oscillator (LSI RC): this is configured by the LSION bit in the Control/status register (RCC_CSR).
- External 32.768 kHz oscillator (LSE OSC): this is configured by the LSEON bit in the RTC domain control register (RCC_BDCR)

Exiting Standby mode

The microcontroller exits the Standby mode when an external reset (NRST pin), an IWDG reset, a rising edge on the WKUP pin or the rising edge of an RTC alarm occurs (see [Figure 294: RTC block diagram](#)). All registers are reset after wake-up from Standby except for [Power control/status register \(PWR_CSR\)](#).

After waking up from Standby mode, program execution restarts in the same way as after a Reset (boot pins sampling, vector reset is fetched, etc.). The SBF status flag in the [Power control/status register \(PWR_CSR\)](#) indicates that the MCU was in Standby mode.

Refer to [Table 21](#) for more details on how to exit Standby mode.

Table 21. Standby mode

Standby mode	Description
Mode entry	WFI (Wait for Interrupt) or WFE (Wait for Event) while: – Set SLEEPDEEP in Cortex-M4 System Control register – Set PDDS bit in Power Control register (PWR_CR) – Clear WUF bit in Power Control/Status register (PWR_CSR)
Mode exit	WKUP pin rising edge, RTC alarm event's rising edge, external Reset in NRST pin, IWDG Reset.
Wakeup latency	Reset phase

I/O states in Standby mode

In Standby mode, all I/O pins are high impedance except:

- Reset pad (still available)
- TAMPER pin if configured for tamper or calibration out
- WKUP pin, if enabled

Debug mode

By default, the debug connection is lost if the application puts the MCU in Stop or Standby mode while the debug features are used. This is because the Arm® Cortex®-M4 core is no longer clocked.

However, by setting some configuration bits in the DBGMCU_CR register, the software can be debugged even when using the low-power modes extensively.

7.3.6 Auto-wakeup from low-power mode

The RTC can be used to wake-up the MCU from low-power mode without depending on an external interrupt (Auto wake-up mode). The RTC provides a programmable time base for waking up from Stop or Standby mode at regular intervals. For this purpose, two of the three alternative RTC clock sources can be selected by programming the RTCSEL[1:0] bits in the [RTC domain control register \(RCC_BDCR\)](#):

- Low-power 32.768 kHz external crystal oscillator (LSE OSC).
This clock source provides a precise time base with very low-power consumption (less than 1µA added consumption in typical conditions)
- Low-power internal RC Oscillator (LSI RC)
This clock source has the advantage of saving the cost of the 32.768 kHz crystal. This internal RC Oscillator is designed to add minimum power consumption.

To wake-up from Stop mode with an RTC alarm event, it is necessary to:

- Configure the EXTI Line 17 to be sensitive to rising edge
- Configure the RTC to generate the RTC alarm

To wake-up from Standby mode, there is no need to configure the EXTI Line 17.

7.4 Power control registers

The peripheral registers can be accessed by half-words (16-bit) or words (32-bit).

7.4.1 Power control register (PWR_CR)

Address offset: 0x00

Reset value: 0x0000 0000 (reset by wake-up from Standby mode)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	DBP	PLS[2:0]			PVDE	CSBF	CWUF	PDDS	LPDS
							rw	rw	rw	rw	rw	rc_w1	rc_w1	rw	rw

Bits 31:9 Reserved, must be kept at reset value.

Bit 8 **DBP**: Disable RTC domain write protection.

In reset state, the RTC and backup registers are protected against parasitic write access. This bit must be set to enable write access to these registers.

0: Access to RTC and Backup registers disabled

1: Access to RTC and Backup registers enabled

Note: If the HSE divided by 128 is used as the RTC clock, this bit must remain set to 1.

Bits 7:5 **PLS[2:0]**: PVD level selection.

These bits are written by software to select the voltage threshold detected by the programmable voltage detector.

000: 2.2V

001: 2.3V

010: 2.4V

011: 2.5V

100: 2.6V

101: 2.7V

110: 2.8V

111: 2.9V

Notes:

1. Refer to the electrical characteristics of the datasheet for more details.
2. Once the PVD_LOCK is enabled (for CLASS B protection) the PLS[2:0] bits cannot be programmed anymore.

Bit 4 **PVDE**: Programmable voltage detector enable.

This bit is set and cleared by software.

0: PVD disabled

1: PVD enabled

Bit 3 **CSBF**: Clear standby flag.

This bit is always read as 0.

0: No effect

1: Clear the SBF Standby Flag (write).

Bit 2 **CWUF**: Clear wakeup flag.

This bit is always read as 0.

0: No effect

1: Clear the WUF Wakeup Flag **after 2 System clock cycles**. (write)

Bit 1 **PDDS**: Power down deepsleep.

This bit is set and cleared by software. It works together with the LPDS bit.

0: Enter Stop mode when the CPU enters Deepsleep. The regulator status depends on the LPDS bit.

1: Enter Standby mode when the CPU enters Deepsleep.

Bit 0 **LPDS**: Low-power deepsleep.

This bit is set and cleared by software. It works together with the PDDS bit.

0: Voltage regulator on during Stop mode

1: Voltage regulator in low-power mode during Stop mode

7.4.2 Power control/status register (PWR_CSR)

Address offset: 0x04

Reset value: 0x0000 0000 (not reset by wake-up from Standby mode)

Additional APB cycles are needed to read this register versus a standard APB read.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	EWUP3 (1)	EWUP2	EWUP1	Res.	Res.	Res.	Res.	VREFIN TRDYF	PVDO	SBF	WUF
					rw	rw	rw					r	r	r	r

1. Not available on STM32F303x6/8 and STM32F328x8.

Bits 31:11 Reserved, must be kept at reset value.

Bit 10 **EWUP3**: Enable WKUP3 pin

This bit is set and cleared by software.

0: WKUP3 pin is used for general purpose I/O. An event on the WKUP3 pin does not wakeup the device from Standby mode.

1: WKUP3 pin is used for wakeup from Standby mode and forced in input pull down configuration (rising edge on WKUP3 pin wakes-up the system from Standby mode).

Note: This bit is reset by a system Reset.

Bit 9 **EWUP2**: Enable WKUP2 pin

This bit is set and cleared by software.

0: WKUP2 pin is used for general purpose I/O. An event on the WKUP2 pin does not wakeup the device from Standby mode.

1: WKUP2 pin is used for wakeup from Standby mode and forced in input pull down configuration (rising edge on WKUP2 pin wakes-up the system from Standby mode).

Note: This bit is reset by a system Reset.

Bit 8 **EWUP1**: Enable WKUP1 pin

This bit is set and cleared by software.

0: WKUP1 pin is used for general purpose I/O. An event on the WKUP1 pin does not wakeup the device from Standby mode.

1: WKUP1 pin is used for wakeup from Standby mode and forced in input pull down configuration (rising edge on WKUP1 pin wakes-up the system from Standby mode).

Note: This bit is reset by a system Reset.

Bits 7:4 Reserved, must be kept at reset value.

Bit 3 **VREFINTRDYF**: V_{REFINT} Ready. Read Only. This bit indicates the state of the internal reference voltage. It is set when V_{REFINT} is ready. It is reset during stabilization of V_{REFINT} .

Note: This flag is useful only for the product bypassing the internal regulator and using external NPOR Pin, the internal POR waits the V_{REFINT} stabilization before releasing the reset.

Bit 2 **PVDO**: PVD output

This bit is set and cleared by hardware. It is valid only if PVD is enabled by the PVDE bit.

0: V_{DD}/V_{DDA} is higher than the PVD threshold selected with the PLS[2:0] bits.

1: V_{DD}/V_{DDA} is lower than the PVD threshold selected with the PLS[2:0] bits.

Notes:

1. The PVD is stopped by Standby mode. For this reason, this bit is equal to 0 after Standby or reset until the PVDE bit is set.
2. Once the PVD is enabled and configured in the PWR_CR register, PVDO can be used to generate an interrupt through the External Interrupt controller.
3. Once the PVD_LOCK is enabled (for CLASS B protection) PVDO cannot be disabled anymore.

Bit 1 **SBF**: Standby flag

This bit is set by hardware and cleared only by a POR/PDR (power on reset/power down reset) or by setting the CSBF bit in the [Power control register \(PWR_CR\)](#)

0: Device has not been in Standby mode

1: Device has been in Standby mode

Bit 0 **WUF**: Wakeup flag

This bit is set by hardware and cleared by a system reset or by setting the CWUF bit in the [Power control register \(PWR_CR\)](#)

0: No wakeup event occurred

1: A wakeup event was received from the WKUP pin or from the RTC alarm

Note: An additional wakeup event is detected if the WKUP pin is enabled (by setting the EWUP bit) when the WKUP pin level is already high.

7.4.3 PWR register map

Table 22. PWR register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x000	PWR_CR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DBP	PLS[2:0]			PVDE	CSBF	CWUF	PDDS	LPDS
	Reset value																								0	0	0	0	0	0	0	0	0

Table 22. PWR register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x004	PWR_CSR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	EWUP3	EWUP2	EWUP1	Res.	Res.	Res.	Res.	VREFINTRDYF	PVDO	SBF	WUF
	Reset value																						0	0	0					0	0	0	0

Refer to [Section 3.2 on page 53](#) for the register boundary addresses.